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Find the derivative of the function.

$$q = \sqrt[3]{x^8 + 9x}$$

First rewrite the function using a rational exponent.

$$\sqrt[3]{x^8 + 9x} = (x^8 + 9x)^{\frac{1}{3}}$$

Suppose that u is a differentiable function of x . The Power Chain Rule states that, for any real number n , $\frac{d}{dx}[u]^n = nu^{n-1} \frac{du}{dx}$.Identify u and n .

$$u = x^8 + 9x$$

$$n = \frac{1}{3}$$

Substitute $u = x^8 + 9x$ and $n = \frac{1}{3}$ into the formula for the Power Chain Rule.

$$\begin{aligned} q' &= \frac{d}{dx}[u(x)]^n \\ &= nu^{n-1} \frac{du}{dx} \\ &= \frac{1}{3} (x^8 + 9x)^{\frac{1}{3}-1} \cdot \frac{d}{dx} (x^8 + 9x) \\ &= \frac{1}{3} (x^8 + 9x)^{-\frac{2}{3}} \cdot \frac{d}{dx} (x^8 + 9x) \end{aligned}$$

Find the derivative of $x^8 + 9x$.

$$\begin{aligned} q' &= \frac{1}{3} (x^8 + 9x)^{-\frac{2}{3}} \cdot \frac{d}{dx} (x^8 + 9x) \\ &= \frac{1}{3} (x^8 + 9x)^{-\frac{2}{3}} (8x^7 + 9) \end{aligned}$$

Rewrite this result using a positive exponent.

$$\frac{1}{3} (x^8 + 9x)^{-\frac{2}{3}} (8x^7 + 9) = \frac{8x^7 + 9}{3(x^8 + 9x)^{\frac{2}{3}}}$$

Therefore, $q' = \frac{8x^7 + 9}{3(x^8 + 9x)^{\frac{2}{3}}}$.